

Land Use and Land Cover of World Regions

Introduction

Land use and land cover (LULC) describe two related but distinct concepts. Land cover refers to the physical material present on the surface of the earth, such as forests, grasslands, water bodies, wetlands, bare soil, ice, or built-up surfaces. Land use, on the other hand, refers to how people utilize the land, such as for agriculture, residential settlement, industry, conservation, or recreation. Together, these two concepts help researchers, planners, and policymakers understand how natural landscapes are being transformed by human activity and natural processes across the globe. This document provides a descriptive, region-by-region overview of the dominant land use and land cover patterns found around the world.

North America

North America exhibits a highly diverse land cover profile shaped by its vast latitudinal extent, from the Arctic tundra of northern Canada and Alaska to the subtropical wetlands of Florida. The boreal forest, or taiga, stretches across much of Canada and Alaska, forming one of the largest contiguous forest belts on the planet. Moving southward, temperate forests dominate the eastern United States and the Pacific Northwest, while the central interior is characterized by expansive grasslands and prairies that have largely been converted into intensive agricultural land for the cultivation of corn, wheat, and soybeans. The arid and semi-arid landscapes of the southwestern United States and northern Mexico feature desert scrub and sparse vegetation. Urbanization is heavily concentrated along the coastlines, particularly the northeastern seaboard, the Pacific coast, and the Gulf Coast, where dense metropolitan land use dominates. Agricultural land use is extensive throughout the Midwest and California's Central Valley, the latter being one of the most productive irrigated farming regions in the world.

South America

South America is defined by the Amazon Basin, which contains the largest tropical rainforest on Earth and represents one of the most ecologically significant land cover types globally. This dense evergreen forest cover spans Brazil, Peru, Colombia, and several neighboring countries, though it faces continuous pressure from deforestation driven by cattle ranching, soybean cultivation, and logging. To the south, the Cerrado savanna of Brazil and the Pampas grasslands of Argentina and Uruguay have been extensively converted into cropland and pasture, making this region a major global exporter of soy, beef, and grain. The Andean highlands present a mix of alpine grassland, or páramo, and terraced agricultural land shaped by centuries of indigenous farming practices. Coastal Peru and northern Chile are dominated by arid desert land cover, including the hyper-arid Atacama Desert, while southern Chile and Argentina transition into temperate forests and Patagonian steppe.

Europe

Europe's land cover reflects thousands of years of intensive human modification, with relatively little remaining pristine wilderness compared to other continents. Agricultural

land use dominates much of the continent, particularly the fertile plains of France, Germany, Poland, and Ukraine, where cereal cultivation and mixed farming are widespread. Temperate deciduous and mixed forests persist in patches throughout central and eastern Europe, often managed for timber production or conservation. Scandinavia and parts of Russia within the European landmass retain substantial boreal forest cover. The Mediterranean region, including Spain, Italy, and Greece, is characterized by a combination of scrubland, olive groves, vineyards, and dryland farming adapted to a hot, dry summer climate. Urban and industrial land use is densely concentrated in a corridor stretching from southeastern England through the Benelux countries, western Germany, and into northern Italy, often referred to as Europe's economic core. Alpine regions such as the Alps and Pyrenees feature mountainous land cover with grassland pasture at higher elevations.

Africa

Africa presents an exceptionally varied land cover mosaic due to its size and range of climatic zones. The Sahara Desert dominates much of North Africa as one of the largest hot desert land cover types in the world, characterized by sand dunes, rocky plateaus, and sparse vegetation. South of the Sahara, the Sahel region forms a transitional zone of semi-arid grassland and shrubland that is highly sensitive to changes in rainfall and land degradation. Central Africa contains the Congo Basin, the second largest tropical rainforest in the world after the Amazon, supporting dense evergreen forest cover across the Democratic Republic of the Congo and surrounding countries. East and southern Africa are dominated by extensive savanna grasslands and woodlands, which support both wildlife conservation areas and pastoral land use by livestock-herding communities. Agricultural land use is widespread throughout the continent, ranging from smallholder subsistence farming in rural areas to large-scale commercial plantations of cash crops such as cocoa, coffee, and palm oil in West and Central Africa. Urban land cover is expanding rapidly around major cities such as Lagos, Cairo, Nairobi, and Johannesburg.

Asia

Asia contains the greatest diversity of land cover of any continent, ranging from polar tundra in Siberia to tropical rainforest in Southeast Asia. Siberia is dominated by an immense boreal forest belt, the largest in the world, interspersed with permafrost and tundra in its northernmost reaches. Central Asia features arid and semi-arid steppe and desert landscapes, including the Gobi Desert, historically used for nomadic pastoralism and increasingly for mineral extraction. South Asia, particularly the Indo-Gangetic Plain of India, Pakistan, and Bangladesh, is one of the most intensively farmed agricultural regions on the planet, supporting rice, wheat, and sugarcane cultivation for enormous population densities. Southeast Asia retains significant tropical rainforest cover in Indonesia, Malaysia, and parts of the Philippines, although large areas have been converted to oil palm and rubber plantations. East Asia, including eastern China, Japan, and South Korea, combines mountainous forested terrain with highly urbanized coastal plains and terraced agricultural land dating back centuries. The Tibetan Plateau and the Himalayan region form a distinct high-altitude land cover zone of alpine grassland, rock, and permanent snow and ice.

Oceania

Oceania's land cover is dominated by the Australian continent, where arid and semi-arid desert and shrubland occupy the vast interior, often referred to as the outback. Tropical savanna woodland covers the northern reaches of Australia, while temperate eucalyptus forest and agricultural land, including wheat belts and sheep and cattle grazing land, are concentrated along the southeastern and southwestern coasts where most of the population resides. New Zealand presents a contrasting profile of temperate rainforest, alpine mountain terrain, and extensively managed pastureland used for dairy and sheep farming. The smaller Pacific Island nations feature a mix of tropical rainforest, coastal mangrove ecosystems, and limited agricultural land constrained by small landmasses and volcanic or coral-based terrain.

Antarctica

Antarctica stands apart from all other regions in that it is almost entirely covered by permanent ice sheets and glaciers, making it the least land-use-modified region on Earth. A very small fraction of exposed rock and soil exists along the coastal margins and in the Antarctic Peninsula, where some seasonal vegetation such as mosses and lichens can be found. There is no permanent agricultural, residential, or industrial land use on the continent, and human presence is limited to scientific research stations operated under international agreements that restrict large-scale land modification.

Global Patterns and Concluding Remarks

Across all regions, several broad global patterns emerge. Forest cover remains concentrated in three major belts: the boreal forests of the high northern latitudes, the temperate forests of the mid-latitudes, and the tropical rainforests near the equator. Agricultural land use has expanded significantly over the past century, often at the expense of natural forest and grassland cover, particularly in South America, Africa, and parts of Asia. Urbanization continues to be a growing driver of land cover change worldwide, with built-up areas expanding fastest in rapidly developing regions of Asia and Africa. Arid and semi-arid lands, which cover a substantial portion of the Earth's surface, remain sparsely used for cultivation but are important for grazing and, increasingly, for renewable energy development such as solar power installations. Understanding these regional patterns of land use and land cover is essential for addressing global challenges such as food security, biodiversity conservation, and climate change mitigation.